

4(a)	oscillations (of object) at maximum amplitude	B1
	when driving frequency equals natural frequency (of object)	B1
4(b)(i)	$T = 2\pi / \omega$	C1
	$= 2\pi / 5.0\pi$	A1
	$= 0.40 \text{ s}$	
4(b)(ii)	displacement scale labelled $-1.0, -0.5, (0), 0.5, 1.0$ on the 2 cm tick marks	B1
	t scale labelled $0.2, 0.4, 0.6, 0.8, 1.0, 1.2$ on the 2 cm tick marks	B1
4(b)(iii)	$\phi = 2\pi\Delta t / T$	C1
	$= 2\pi \times 0.10 / 0.40$ or $2\pi \times 0.30 / 0.40$	
	$= 1.6 \text{ rad}$ or 4.7 rad	A1